

Ethics in Networks Research

A Summary Overview

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Note: Learn more about the Network Ethics Working Group, and find any updates at:
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Aims

This report consolidates and outlines recommended approaches for adhering to ethical standards in network research. We focus specifically on elements that are unique to network research, while leaving broader ethical issues to be addressed within each researcher’s home discipline. This resource serves to assist researchers in their interactions with their institution’s ethical review committees and includes linked resources to support these practices. We have organized our recommendations into three sections, which cover: (1) consent and personal data, (2) analysis and presentation, and (3) data storage and sharing.

Consent & Personal Data

Ethical considerations in network research are grounded in the general principles of human subjects’ research, including avoiding harm, promoting justice, equitable distribution of burdens and benefits, respecting dignity, and protecting privacy. However, applying these principles presents challenges because gathering network data involves collecting data on multiple people simultaneously. Traditional practices such as obtaining individual informed consent, ensuring voluntary participation, and maintaining confidentiality can become complicated when information about secondary participants (i.e., individuals who are named as network alters by participants but who do not themselves directly participate in the research) can still be disclosed by others. Additionally, sociocentric study designs may make complete anonymization impossible. To address these challenges, researchers need to adapt standard procedures for obtaining participant consent and ensuring privacy protections, and need to employ pseudonymization or anonymization whenever possible.^{1–3}

To address the ethical complexities of consent for non-participants, researchers have available three primary strategies. The first involves securing consent from entire closed populations in advance, covering both personal participation and information shared by others. The second considers alters as “secondary participants,” and draws on practices from other fields like demographic surveys, where representatives provide information about secondary participants under principles distinguishing private from publicly knowable data. This approach recognizes the need for individual consent may be waived for secondary participants when network research does not violate privacy standards, is minimal risk, and could not otherwise be conducted.⁴ A third strategy is related to how we manage personal data that may be identifiable. Adapted from chain-referral methods, it may be helpful to anonymize nomination and recruitment, enabling relational data to be linkable without directly compromising privacy.

Privacy protections in network research often couple closely with consent, unlike in traditional human subjects’ research where these may be treated separately. Researchers typically aim to anonymize or pseudonymize data, though in relational designs real identifiers may be necessary before later protection steps. Digital trace methods follow similar principles, requiring both ego consent and careful handling of alters’ data. Additional approaches involve minimizing data collection to only what is strictly necessary, offering feedback to participants to encourage retention, and, in some jurisdictions, relying on public interest exceptions that allow limited use of personal data without explicit consent. These adaptations highlight the distinctive ethical terrain of network research, where consent and privacy are inseparably linked.^{5–7}

Data Presentation, Analysis, & Visualization

Network visualization is a valuable tool for offering participants’ feedback and generating new narratives that can provide insights into the data. However, researchers should be sensitive to the

fact that network visualization may lead to easy identification of individual participants. Such identification might highlight individuals who are in vulnerable network positions (e.g., isolates) or emphasize social divides. Therefore, if visualizations are shared with participants or in research reports, researchers should adopt the necessary mitigation measures to avoid unnecessary discomfort, especially when working with vulnerable or culturally diverse populations.⁸

Depending on the type of research conducted, researchers may also pay attention to the visualization, and how visualization choices could also enhance or complicate communication of results. To this end, network researchers should combine visual variables like size, color, position, or shape in an informative way, paying attention to not magnify centrality, isolation, or segregation. When necessary, it is also important to provide alternative visualizations of the same data, along with contextual explanations of their meanings and limitations. These recommendations are equally relevant when conducting participatory research using network diagrams.

To protect the identities of both participants and secondary participants, network visualizations should avoid including labels with personal names or specific geographical information. If contextual information such as age, race, location, or unique network characteristics could potentially reveal identities, it is advisable not to include or use that information in visualizations. Furthermore, in institutional settings, it is crucial to have an agreement with the organization that prevents the display or use of personal network identifiers for individual assessments.^{9–11}

Publication, Data Storage & Sharing

As a fundamental principle, research data should be securely stored and managed in a secure repository throughout the data lifecycle.¹² Researchers must adhere to governmental and institutional regulations, and wherever relevant, comply with disciplinary standards and with publishers' guidelines. For example, researchers working with Indigenous data should adhere to principles of Indigenous data.¹³ These regulations can change over time (for instance, regarding issues like data scraping) and may differ across jurisdictions.

Publications and presentations should adhere to the same principles as outlined for visualization above, avoiding labels or details that could lead to personal identification.¹⁴ In the case of face-to-face interviews, a good practice is to include in the consent form an option about whether the participants agree that verbatim transcriptions of their interviews can be published in an anonymized way. When appropriate, individuals involved may be consulted for feedback on specific portions of the published material that include these transcriptions.

When anonymizing data, personal identifiers should be kept in a separate folder that is restricted to authorized personnel only. Typically, anonymization or de-identification before dataset publication involves substituting identifiable information with codes, consolidating variables, reducing the granularity of existing data, employing broader geographic markers, and other similar techniques.¹⁵

Whenever possible, it is recommended to share the data in an online repository (e.g., GitHub, Open Science Framework–OSF), with the necessary anonymization, pseudonymization, or de-identification.¹² In that case, sharing this data repository with the participant may also be recommended. If it is not possible to share all the data, the author(s) may consider sharing a selected sample or posting materials to OSF.¹⁶ Developing a data management plan and deciding on a way to share the data from the beginning may improve replicability for the scientific community and transparency for participants.

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